LAPORAN PRAKTIKUM 1
SISTEM OPERASI

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BAB I

PENDAHULUAN

LATAR BELAKANG

Sistem operasi merupakan perangkat lunak inti yang mengatur dan mengendalikan seluruh sumber daya komputer, baik perangkat keras maupun perangkat lunak. Tanpa sistem operasi, pengguna tidak dapat berinteraksi dengan komputer secara efektif karena semua proses seperti pengelolaan memori, CPU, penyimpanan, dan perangkat I/O dijalankan serta dikendalikan oleh sistem operasi. Menurut Silberschatz (2020), sistem operasi berperan sebagai pengelola sumber daya (*resource manager*) dan penyedia antarmuka antara pengguna serta perangkat keras.

Melalui praktikum Sistem Operasi ini, mahasiswa mempelajari secara langsung bagaimana sistem operasi bekerja, khususnya dalam lingkungan Linux. Materi yang dipelajari meliputi layanan sistem operasi (*Operating-System Services*), antarmuka pengguna dan sistem (*User and Operating-System Interface*), serta pemanggilan sistem (*System Calls*) yang menjadi sarana komunikasi antara program dan kernel. Selain itu, mahasiswa juga diperkenalkan pada *System Services*, proses *linking* dan *loading*, serta struktur dan desain sistem operasi modern.

Dalam praktikum ini, mahasiswa menerapkan konsep-konsep tersebut melalui pembuatan program sederhana menggunakan bahasa C untuk memahami hubungan antara shell, kernel, dan aplikasi. Melalui kegiatan ini, mahasiswa memperoleh pemahaman yang lebih mendalam tentang cara kerja sistem operasi dalam mengelola sumber daya, menjalankan proses, serta menjaga stabilitas sistem. Dengan demikian, praktikum ini memberikan dasar penting bagi mahasiswa Teknik Informatika dalam memahami mekanisme kerja sistem komputer secara menyeluruh dan efisien.

RUMUSAN MASALAH

1. Bagaimana sistem operasi bekerja dalam mengelola sumber daya komputer seperti proses, memori, dan file?
2. Bagaimana mekanisme antarmuka antara pengguna, shell, dan kernel melalui Command Line Interface (CLI) serta system calls?
3. Bagaimana penerapan konsep layanan sistem operasi dapat dipahami melalui praktik langsung menggunakan program C pada lingkungan Linux?

TUJUAN

1. Untuk memahami fungsi dan cara kerja sistem operasi dalam mengatur serta mengoordinasikan sumber daya komputer.
2. Untuk mempelajari proses komunikasi antara pengguna, shell, dan kernel melalui perintah Linux dan implementasi system calls.
3. Untuk meningkatkan kemampuan praktis mahasiswa dalam mengaplikasikan teori sistem operasi melalui eksperimen dan pemrograman sederhana di Linux.

BAB II

PEMBAHASAN

2.1 Operating-System Services

Layanan sistem operasi (Operating-System Services) adalah fungsi penting yang disediakan OS untuk mendukung program dan pengguna. Praktik ini menunjukkan cara OS bekerja melalui perintah Linux dan program C yang menampilkan layanan seperti eksekusi proses, operasi I/O, manipulasi file, komunikasi, deteksi kesalahan, serta alokasi sumber daya secara efisien.

2.1.1 Menjelajahi Layanan OS

- **Lihat informasi system**

```
asus@LAPTOP-BV04OHF1: ~
asus@LAPTOP-BV04OHF1:~$ uname -a
Linux LAPTOP-BV04OHF1 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
asus@LAPTOP-BV04OHF1:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
```

- **Lihat proses yang berjalan**

```
asus@LAPTOP-BV04OHF1:~$ ps aux | head -20
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.8  0.1  21680 11988 ?        Ss   20:51   0:00 /sbin/init
root           2  0.0  0.0   3072  1664 ?        Sl   20:51   0:00 /init
root           7  0.0  0.0   3072  1792 ?        Sl   20:51   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          41  0.2  0.1  42164 15744 ?        S<s  20:51   0:00 /usr/lib/systemd/systemd-journald
root          91  0.2  0.0  25092  6272 ?        Ss   20:51   0:00 /usr/lib/systemd/systemd-udevd
systemd+     112  0.1  0.1  21456 12544 ?        Ss   20:51   0:00 /usr/lib/systemd/systemd-resolved
systemd+     116  0.1  0.0  91024  7424 ?        Ssl  20:51   0:00 /usr/lib/systemd/systemd-timesyncd
root         177  0.0  0.0   4236  2560 ?        Ss   20:51   0:00 /usr/sbin/cron -f -P
message+     178  0.0  0.0   9592  4736 ?        Ss   20:51   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
og-only
root         185  0.1  0.1  17960  8192 ?        Ss   20:51   0:00 /usr/lib/systemd/systemd-logind
root         187  0.1  0.1 1756096 12032 ?       Ssl  20:51   0:00 /usr/libexec/wsl-pro-service -vv
root         197  0.0  0.0   3160  1920 hvc0     Ss+  20:51   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root         210  0.0  0.0   3116  1792 tty1     Ss+  20:51   0:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog       212  0.1  0.0 222508  5504 ?        Ssl  20:51   0:00 /usr/sbin/rsyslogd -n -iNONE
root         218  0.1  0.2 107032 22656 ?        Ssl  20:51   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root         307  0.0  0.0   3076  1024 ?        Ss   20:51   0:00 /init
root         308  0.0  0.0   3092  1028 ?        S    20:51   0:00 /init
asus         309  0.0  0.0   6072  5248 pts/0    Ss   20:51   0:00 -bash
root         310  0.0  0.0   6820  4096 pts/1    Ss   20:51   0:00 /bin/login -f
```

- **Lihat penggunaan memori**

```
asus@LAPTOP-BV04OHF1:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.7Gi       497Mi       6.9Gi       3.5Mi       407Mi       7.2Gi
Swap:          2.0Gi          0B       2.0Gi
```

- **Lihat informasi CPU**

```
asus@LAPTOP-BV040HF1:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          48 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   16
  On-line CPU(s) list:    0-15
Vendor ID:                AuthenticAMD
  Model name:             AMD Ryzen 7 7435HS
    CPU family:           25
    Model:                68
    Thread(s) per core:   2
    Core(s) per socket:   8
    Socket(s):            1
    Stepping:             1
    BogoMIPS:             6188.19
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse s
                          se2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable no
                          nstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe po
                          pcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalign
                          sse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 sm
                          ep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clz
                          ero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter p
                          fthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
Virtualization features:
  Virtualization:         AMD-V
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    256 KiB (8 instances)
  L1i:                    256 KiB (8 instances)
  L2:                     4 MiB (8 instances)
  L3:                     16 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-15
Vulnerabilities:
  Gather data sampling:   Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Reg file data sampling: Not affected
  Retbleed:               Not affected
  Spec rstack overflow:   Vulnerable: Safe RET, no microcode
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSB-eIBRS N
                          ot affected; BHI Not affected
  Srbds:                  Not affected
  Tsx async abort:        Not affected
```

- **Lihat perangkat yang terpasang**

```
asus@LAPTOP-BV040HF1:~$ lsblk
NAME
    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.4M  1 disk
sdb   8:16   0   186M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
                              /
```

2.1.2 Program C - Menggunakan Layanan OS

```
asus@LAPTOP-BV040HF1:~$ nano os_services.c
asus@LAPTOP-BV040HF1:~$ gcc -o os_services os_services.c
asus@LAPTOP-BV040HF1:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 491
Parent Process ID (PPID): 309

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
```

2.2 User and Operating-System Interface

User and Operating-System Interface adalah mekanisme yang memungkinkan pengguna berinteraksi dengan sistem operasi. Melalui Command Line Interface (CLI), pengguna dapat menjalankan perintah seperti navigasi direktori, penggunaan pipelining, redirection, serta menjalankan proses di latar belakang. Pada praktik simple_shell.c, dibuat shell sederhana yang menerima input pengguna, memecah argumen, dan mengeksekusi perintah menggunakan fungsi fork() dan execvp(). Praktik ini memperlihatkan bagaimana antarmuka antara pengguna, shell, dan kernel bekerja untuk menjalankan instruksi secara langsung dan efisien dalam sistem operasi Linux.

2.2.1 Command Line Interface (CLI)

- **Navigasi dasar**

```
asus@ Default: asus@LAPTOP-BV04OHF1: /tmp/test_dir
/home        ctrl+alt+1
asus@LAPTOP-BV04OHF1:~$ ls -la
total 1064
drwxr-x--- 7 asus asus   4096 Oct 13 21:17  .
drwxr-xr-x 3 root root   4096 Oct  7 21:22  ..
-rw------- 1 asus asus   6571 Oct 13 19:30  .bash_history
-rw-r--r-- 1 asus asus    220 Oct  7 21:22  .bash_logout
-rw-r--r-- 1 asus asus   3771 Oct  7 21:22  .bashrc
drwx------ 2 asus asus   4096 Oct  7 21:24  .cache
drwx------ 3 asus asus   4096 Oct 10 17:47  .config
-rw------- 1 asus asus     20 Oct 10 21:06  .lesshst
drwxr-xr-x 3 asus asus   4096 Oct  7 21:51  .local
-rw-rw-r-- 1 asus asus      0 Oct 13 18:33  .motd_shown
-rw-r--r-- 1 asus asus    807 Oct  7 21:22  .profile
-rw-r--r-- 1 asus asus      0 Oct  7 22:11  .sudo_as_admin_successful
-rw-r--r-- 1 root root    178 Oct  9 19:35 'Description=My'
-rwxr-xr-x 1 asus asus  17400 Oct 12 19:43  file_monitor
-rw-r--r-- 1 asus asus   7174 Oct 12 19:42  file_monitor.c
-rw-r--r-- 1 asus asus    448 Oct 13 18:33  hello_module.c
-rwxr-xr-x 1 asus asus  15144 Oct 10 16:47  libmathops.so
-rw-r--r-- 1 asus asus    154 Oct 10 12:05  main.c
-rw-r--r-- 1 asus asus  21452 Oct 10 12:05  main.i
-rw-r--r-- 1 asus asus   1728 Oct 10 12:08  main.o
-rw-r--r-- 1 asus asus    910 Oct 10 12:07  main.s
-rw-r--r-- 1 root root    166 Oct 10 17:11  makefile
-rw-r--r-- 1 asus asus    118 Oct 10 12:03  math_ops.c
-rw-r--r-- 1 asus asus     98 Oct 10 11:59  math_ops.h
-rw-r--r-- 1 asus asus    322 Oct 10 12:05  math_ops.i
-rw-r--r-- 1 asus asus   1352 Oct 10 16:47  math_ops.o
-rw-r--r-- 1 asus asus    992 Oct 10 12:07  math_ops.s
-rw-r--r-- 1 asus asus    785 Oct  9 19:30  my_daemon.c
-rwxr-xr-x 1 asus asus  16488 Oct 13 21:06  os_services
-rw-r--r-- 1 asus asus   1449 Oct 13 21:06  os_services.c
-rw-r--r-- 1 asus asus     53 Oct  7 21:59  os_services.c:
-rw-r--r-- 1 asus asus    761 Oct 10 16:59  os_specific.c
drwxr-xr-x 5 asus asus   4096 Oct 12 19:55  praktikum_os
-rwxr-xr-x 1 asus asus  17040 Oct 11 15:40  process_manager
-rw-r--r-- 1 asus asus   3785 Oct 11 15:40  process_manager.c
-rwxr-xr-x 1 asus asus  16256 Oct 10 21:00  procfs_read
-rw-r--r-- 1 asus asus   1175 Oct 10 21:00  procfs_read.c
-rwxr-xr-x 1 asus asus  16056 Oct 10 12:08  program
-rwxr-xr-x 1 asus asus  16056 Oct 10 16:42  program_dynamic
-rwxr-xr-x 1 asus asus  16024 Oct 10 16:48  program_shared
-rwxr-xr-x 1 asus asus 785400 Oct 10 16:42  program_static
drwxr-xr-x 2 asus asus   4096 Oct 13 21:17  test_dir
-rw-r--r-- 1 asus asus     23 Oct 13 21:06  test_file.txt
asus@LAPTOP-BV04OHF1:~$ cd /tmp
asus@LAPTOP-BV04OHF1:/tmp$ mkdir test_dir
asus@LAPTOP-BV04OHF1:/tmp$ cd test_dir
```

- **Pipelining**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ps aux | grep bash | wc -l
3
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

- **Redirection**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ echo "Hello World" > output.txt
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat output.txt
Hello World
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat < output.txt
Hello World
```

- **Background processes**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sleep 30 &
[1] 1005
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ jobs
[1]+  Running                 sleep 30 &
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ fg %1
sleep 30
```

2.2.2 Program C - Shell Sederhana

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano simple_shell.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -o simple_shell simple_shell.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ./simple_shell
myshell> echo Hallo world
Hallo world
myshell> ^Z
[1]+  Stopped                 ./simple_shell
```

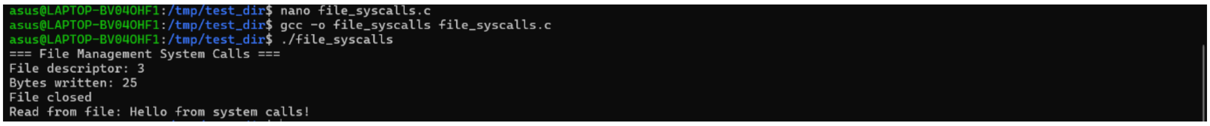
2.3 System Calls

System Calls adalah mekanisme yang digunakan program untuk berkomunikasi langsung dengan kernel sistem operasi. Dalam praktik ini, pengguna mempelajari tiga jenis system calls utama. Pertama, Process Control System Calls seperti fork(), exec(), dan wait() digunakan untuk membuat, menjalankan, dan mengelola proses. Kedua, File Management System Calls seperti open(), write(), read(), dan close() digunakan untuk mengelola file pada sistem. Ketiga, Information Maintenance System Calls seperti getpid(), getuid(), dan time() digunakan untuk memperoleh informasi tentang proses dan waktu sistem. Melalui praktik ini, pengguna memahami bagaimana sistem operasi menyediakan layanan penting untuk pengelolaan proses, file, dan informasi sistem secara efisien.

2.3.1 Process Control System Calls

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano process_syscalls.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -o process_syscalls process_syscalls.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 480
Parent waiting for child (PID: 481)...
Child PID: 481
Child's Parent PID: 480
Child executing 'ls' command...
total 44
-rwxr-xr-x 1 asus asus 16360 Oct 13 22:04 process_syscalls
-rw-r--r-- 1 asus asus   798 Oct 13 22:04 process_syscalls.c
-rwxr-xr-x 1 asus asus 16528 Oct 13 21:48 simple_shell
-rw-r--r-- 1 asus asus  1190 Oct 13 21:48 simple_shell.c
Child exited with status: 0
```

2.3.2 File Management System Calls



```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano file_syscalls.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -o file_syscalls file_syscalls.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
```

2.3.3 Information Maintenance System Calls

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano info_syscalls.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -o info_syscalls info_syscalls.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 509
Parent Process ID: 284
User ID: 1000
Group ID: 1000
Current time: Mon Oct 13 22:16:17 2025
Clock ticks since boot: 1718456236
Sleeping for 2 seconds...
Awake!
```

2.4 System Services

System Services di Linux adalah proses yang berjalan di latar belakang untuk menyediakan berbagai fungsi penting seperti jaringan, keamanan, dan manajemen sistem. Dalam praktik ini, digunakan systemd sebagai sistem inisialisasi modern untuk mengelola layanan. Dengan perintah seperti systemctl list-units, pengguna dapat melihat daftar layanan aktif, memeriksa statusnya menggunakan systemctl status, dan melihat log aktivitas melalui journalctl. Selain itu, dibuat program daemon sederhana (my_daemon.c) yang berjalan otomatis di latar belakang dan mencatat aktivitas ke syslog. Pembuatan file service di /etc/systemd/system/ memungkinkan daemon tersebut dikelola oleh systemd. Praktik ini membantu memahami bagaimana sistem operasi menjalankan, memantau, dan mengontrol layanan secara efisien serta berkesinambungan.

2.4.1 Menggunakan system

- **Lihat semua service**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ systemctl list-units --type=service
  UNIT                                 LOAD   ACTIVE SUB     DESCRIPTION
  console-getty.service                loaded active running Console Getty
  console-setup.service                loaded active exited  Set console font and keymap
  cron.service                         loaded active running Regular background program processing daemon
  dbus.service                         loaded active running D-Bus System Message Bus
  getty@tty1.service                   loaded active running Getty on tty1
  keyboard-setup.service               loaded active exited  Set the console keyboard layout
  kmod-static-nodes.service            loaded active exited  Create List of Static Device Nodes
  rsyslog.service                      loaded active running System Logging Service
  setvtrgb.service                     loaded active exited  Set console scheme
  snapd.seeded.service                 loaded active exited  Wait until snapd is fully seeded
  systemd-binfmt.service               loaded active exited  Set Up Additional Binary Formats
  systemd-journal-flush.service        loaded active exited  Flush Journal to Persistent Storage
  systemd-journald.service             loaded active running Journal Service
  systemd-logind.service               loaded active running User Login Management
  systemd-modules-load.service         loaded active exited  Load Kernel Modules
  systemd-remount-fs.service           loaded active exited  Remount Root and Kernel File Systems
  systemd-resolved.service             loaded active running Network Name Resolution
  systemd-sysctl.service               loaded active exited  Apply Kernel Variables
  systemd-timesyncd.service            loaded active running Network Time Synchronization
  systemd-tmpfiles-setup-dev-early.service loaded active exited  Create Static Device Nodes in /dev gracefully
  systemd-tmpfiles-setup-dev.service   loaded active exited  Create Static Device Nodes in /dev
  systemd-tmpfiles-setup.service       loaded active exited  Create Volatile Files and Directories
  systemd-udev-trigger.service         loaded active exited  Coldplug All udev Devices
  systemd-udevd.service                loaded active running Rule-based Manager for Device Events and Files
  systemd-update-utmp.service          loaded active exited  Record System Boot/Shutdown in UTMP
  systemd-user-sessions.service        loaded active exited  Permit User Sessions
  unattended-upgrades.service          loaded active running Unattended Upgrades Shutdown
  user-runtime-dir@1000.service        loaded active exited  User Runtime Directory /run/user/1000
  user@1000.service                    loaded active running User Manager for UID 1000
  wsl-pro.service                      loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

30 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
```

- **Status service**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: inactive (dead)
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
```

- **Lihat log service**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ journalctl -u ssh -n 20
Oct 09 18:55:00 LAPTOP-BV04OHF1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 09 18:55:00 LAPTOP-BV04OHF1 sshd[3041]: Server listening on 0.0.0.0 port 22.
Oct 09 18:55:00 LAPTOP-BV04OHF1 sshd[3041]: Server listening on :: port 22.
Oct 09 18:55:00 LAPTOP-BV04OHF1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

- **Lihat service yang berjalan saat boot**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                              STATE   PRESET
apparmor.service                       enabled enabled
apport.service                         enabled enabled
cloud-config.service                   enabled enabled
cloud-final.service                    enabled enabled
cloud-init-local.service               enabled enabled
cloud-init.service                     enabled enabled
console-setup.service                  enabled enabled
cron.service                           enabled enabled
dmesg.service                          enabled enabled
e2scrub_reap.service                   enabled enabled
getty@.service                         enabled enabled
keyboard-setup.service                 enabled enabled
landscape-client.service               enabled enabled
networkd-dispatcher.service            enabled enabled
rsyslog.service                        enabled enabled
setvtrgb.service                       enabled enabled
snapd.apparmor.service                 enabled enabled
snapd.autoimport.service               enabled enabled
snapd.core-fixup.service               enabled enabled
snapd.recovery-chooser-trigger.service enabled enabled
snapd.seeded.service                   enabled enabled
snapd.service                          enabled enabled
snapd.system-shutdown.service          enabled enabled
systemd-pstore.service                 enabled enabled
systemd-resolved.service               enabled enabled
systemd-timesyncd.service              enabled enabled
ua-reboot-cmds.service                 enabled enabled
ubuntu-advantage.service               enabled enabled
unattended-upgrades.service            enabled enabled
wsl-pro.service                        enabled enabled

30 unit files listed.
```

2.4.2 Membuat Custom System Service

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano my_daemon.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -o my_daemon my_daemon.c
```

```
asus@LAPTOP-BV04OHF1: /tn
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo nano /etc/systemd/system/my_daemon.service
[sudo] password for asus:
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo systemctl daemon-reload
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo systemctl start my_daemon
Job for my_daemon.service failed because the control process exited with error code.
See "systemctl status my_daemon.service" and "journalctl -xeu my_daemon.service" for details.
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo systemctl status my_daemon
× my_daemon.service - My Custom Daemon
     Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
     Active: failed (Result: exit-code) since Mon 2025-10-13 22:51:31 WIB; 5s ago
    Process: 659 ExecStart=/path/to/my_daemon (code=exited, status=203/EXEC)
        CPU: 2ms

Oct 13 22:51:31 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 13 22:51:31 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 13 22:51:31 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 13 22:51:31 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo journalctl -u my_daemon
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:40:56 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 4.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:40:57 LAPTOP-BV04OHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 19:43:11 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:43:12 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 19:43:12 LAPTOP-BV04OHF1 systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 09 19:43:12 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 19:43:12 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:43:12 LAPTOP-BV04OHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
lines 1-40
[2]+  Stopped                 sudo journalctl -u my_daemon
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo systemctl stop my_daemon
```

2.5 Linkers and Loaders

Pada Linux, *Linkers and Loaders* berperan penting dalam proses pembuatan program dari kode sumber menjadi file eksekusi. Linker bertugas menggabungkan beberapa file object (*.o*) seperti main.o dan math_ops.o menjadi satu program utuh, sedangkan loader memuat program tersebut ke memori saat dijalankan. Proses kompilasi dilakukan bertahap: preprocessing, compilation, assembly, dan linking. Dalam praktik ini, dibuat fungsi add() dan multiply() yang dihubungkan melalui file header math_ops.h. Selain itu, dibedakan antara *static linking* (semua library disertakan langsung dalam program) dan *dynamic linking* (library eksternal digunakan saat runtime). Shared library (libmathops.so) memungkinkan efisiensi memori dan kemudahan pemeliharaan karena dapat digunakan oleh banyak program sekaligus.

2.5.1 Proses Linking

- **Buat File math_ops.h**

- **Buat File math_ops.c**

- **Buat File main.c**

- **Preprocessing**

- **Compilation (assembly)**

- **Assembly (object files)**

- **Linking**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc main.o math_ops.o -o program
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- ○ **Lihat symbols dalam object file**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nm main.o
                 U add
0000000000000000 T main
                 U multiply
                 U printf
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
                 w _ITM_deregisterTMCloneTable
                 w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
0000000000004010 B __bss_start
                 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
                 w __gmon_start__
                 U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
                 U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- ○ **Lihat dependencies**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ldd program
        linux-vdso.so.1 (0x00007fffa24e7000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000071e460c00000)
        /lib64/ld-linux-x86-64.so.2 (0x000071e460f7c000)
```

2.5.2 Static vs Dynamic Linking

- **Static linking**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc main.c math_ops.c -o program_static -static
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Dynamic linking (default)**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc main.c math_ops.c -o program_dynamic
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Bandingkan ukuran**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 asus asus  16K Oct 13 23:16 program_dynamic
-rwxr-xr-x 1 asus asus 767K Oct 13 23:15 program_static
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat dependencies**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 asus asus  16K Oct 13 23:16 program_dynamic
-rwxr-xr-x 1 asus asus 767K Oct 13 23:15 program_static
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ldd program_static
        not a dynamic executable
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ldd program_dynamic
        linux-vdso.so.1 (0x00007ffe6a94e000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000075503dc00000)
        /lib64/ld-linux-x86-64.so.2 (0x000075503df42000)
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

2.5.3 Membuat dan Menggunakan Shared Library

- **Compile sebagai shared library**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -c -fPIC math_ops.c -o math_ops.o
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -shared -o libmathops.so math_ops.o
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

- **Compile program yang menggunakan library**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc main.c -L. -lmathops -o program_shared
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

- **Jalankan dengan LD_LIBRARY_PATH**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

- **Lihat informasi loader**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
  Tag        Type                         Name/Value
 0x0000000000000001 (NEEDED)             Shared library: [libmathops.so]
 0x0000000000000001 (NEEDED)             Shared library: [libc.so.6]
 0x000000000000000c (INIT)               0x1000
 0x000000000000000d (FINI)               0x11e4
 0x0000000000000019 (INIT_ARRAY)         0x3d98
 0x000000000000001b (INIT_ARRAYSZ)       8 (bytes)
 0x000000000000001a (FINI_ARRAY)         0x3da0
 0x000000000000001c (FINI_ARRAYSZ)       8 (bytes)
 0x000000006ffffef5 (GNU_HASH)           0x3b0
 0x0000000000000005 (STRTAB)             0x4b0
 0x0000000000000006 (SYMTAB)             0x3d8
 0x000000000000000a (STRSZ)              170 (bytes)
 0x000000000000000b (SYMENT)             24 (bytes)
 0x0000000000000015 (DEBUG)              0x0
 0x0000000000000003 (PLTGOT)             0x3fa8
 0x0000000000000002 (PLTRELSZ)           72 (bytes)
 0x0000000000000014 (PLTREL)             RELA
 0x0000000000000017 (JMPREL)             0x660
 0x0000000000000007 (RELA)               0x5a0
 0x0000000000000008 (RELASZ)             192 (bytes)
 0x0000000000000009 (RELAENT)            24 (bytes)
 0x000000000000001e (FLAGS)              BIND_NOW
 0x000000006ffffffb (FLAGS_1)            Flags: NOW PIE
 0x000000006ffffffe (VERNEED)            0x570
 0x000000006fffffff (VERNEEDNUM)         1
 0x000000006ffffff0 (VERSYM)             0x55a
 0x000000006ffffff9 (RELACOUNT)          3
 0x0000000000000000 (NULL)               0x0
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

2.6 Why Applications Are Operating-System Specific

Aplikasi bersifat spesifik terhadap sistem operasi karena setiap OS memiliki *system call*, format biner, dan API (Application Programming Interface) yang berbeda. Pada praktik ini, program os_specific.c menunjukkan bagaimana kode C harus disesuaikan untuk Linux dan Windows. Misalnya, di Linux digunakan pustaka <unistd.h> dan fungsi uname() untuk mendapatkan informasi sistem, sedangkan di Windows menggunakan <windows.h>. Selain itu, perbedaan format biner seperti ELF di Linux dan PE di Windows menyebabkan program tidak dapat dijalankan lintas sistem. Dengan perintah seperti readelf dan objdump, pengguna dapat menganalisis struktur file eksekusi dan memahami bagaimana sistem operasi memuat serta menjalankan aplikasi tersebut.

2.6.1 System Call Differences

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano os_specific.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

2.6.2 Binary Format Analysis

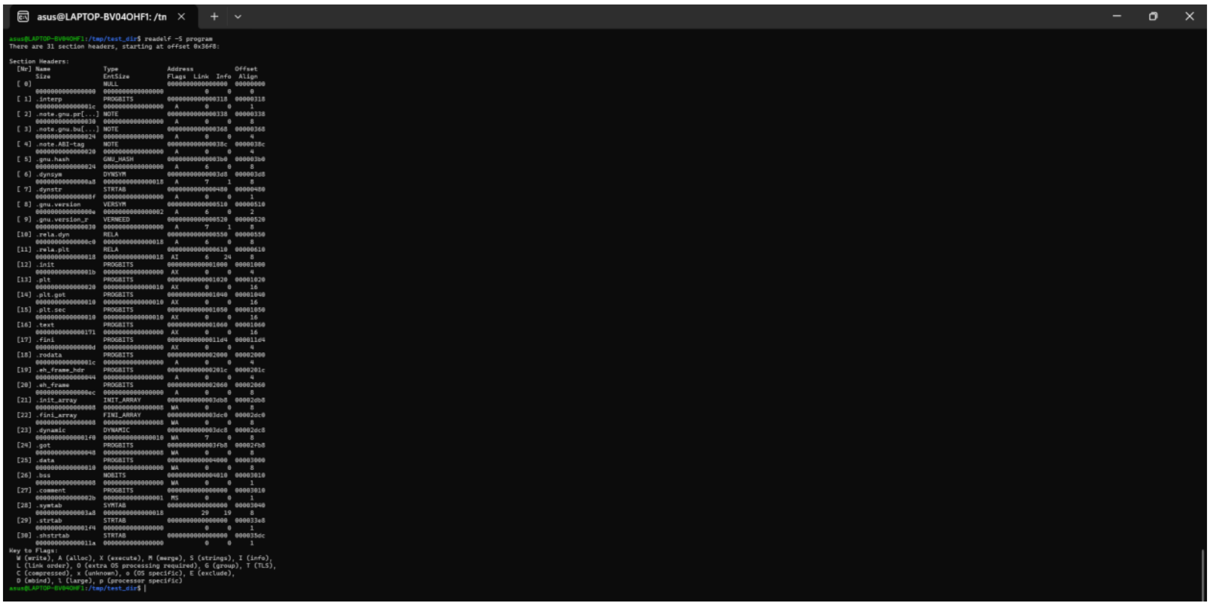
- **Lihat format executable**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b47095af229368edc342f23076fc2, for G
NU/Linux 3.2.0, not stripped
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

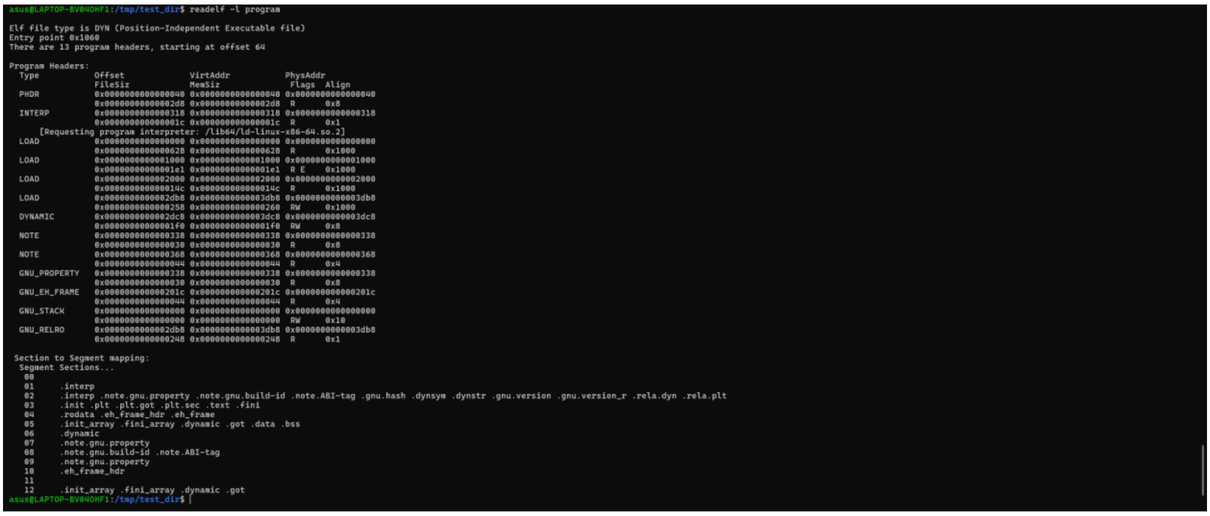
- **Lihat header ELF**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:                             ELF64
  Data:                              2's complement, little endian
  Version:                           1 (current)
  OS/ABI:                            UNIX - System V
  ABI Version:                       0
  Type:                              DYN (Position-Independent Executable file)
  Machine:                           Advanced Micro Devices X86-64
  Version:                           0x1
  Entry point address:               0x1060
  Start of program headers:          64 (bytes into file)
  Start of section headers:          14072 (bytes into file)
  Flags:                             0x0
  Size of this header:               64 (bytes)
  Size of program headers:           56 (bytes)
  Number of program headers:         13
  Size of section headers:           64 (bytes)
  Number of section headers:         31
  Section header string table index: 30
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat section headers**



- **Lihat program headers**



- **Disassemble**

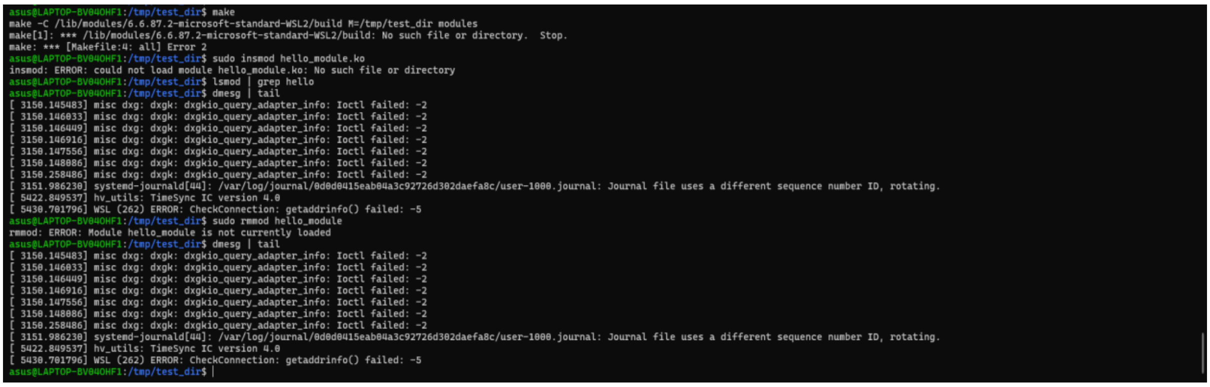
2.7 OS Design and Implementation

Pada praktik ini, mahasiswa mempelajari bagaimana desain dan implementasi sistem operasi dilakukan melalui pembuatan *kernel module* sederhana di Linux. Modul kernel adalah komponen yang dapat dimuat dan dilepas dari kernel secara dinamis tanpa perlu melakukan reboot sistem. Pada file hello_module.c, digunakan fungsi printk() untuk menampilkan pesan ke *kernel log* saat modul dimuat (insmod) dan dilepas (rmmod). Proses kompilasi dilakukan menggunakan *Makefile* yang memanggil build system kernel melalui make -C /lib/modules/.... Selain itu, perintah seperti lsmod, dmesg, dan modinfo digunakan untuk memantau aktivitas kernel dan modul yang aktif. Praktik ini membantu memahami struktur internal kernel serta interaksi antara modul dan sistem operasi.

2.7.1 Kernel Module Sederhana

- **Makefile**

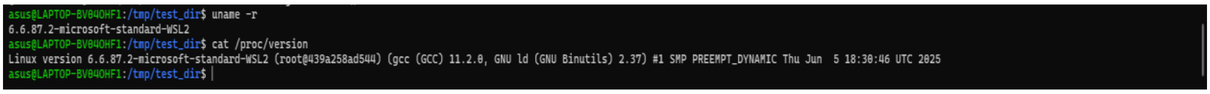
- **Compile and Load**



```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ make
make -C /lib/modules/6.6.87.2-microsoft-standard-WSL2/build M=/tmp/test_dir modules
make[1]: *** /lib/modules/6.6.87.2-microsoft-standard-WSL2/build: No such file or directory.  Stop.
make: *** [Makefile:4: all] Error 2
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ lsmod | grep hello
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ dmesg | tail
[ 3150.145483] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146033] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146449] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.147556] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.148086] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.258486] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3151.986230] systemd-journald[44]: /var/log/journal/0d0d0415eab04a3c92726d302daefa8c/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 5422.849537] hv_utils: TimeSync IC version 4.0
[ 5430.701796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -5
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ dmesg | tail
[ 3150.145483] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146033] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146449] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.147556] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.148086] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.258486] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3151.986230] systemd-journald[44]: /var/log/journal/0d0d0415eab04a3c92726d302daefa8c/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 5422.849537] hv_utils: TimeSync IC version 4.0
[ 5430.701796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -5
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

2.7.2 Monitoring Kernel

- **Lihat informasi kernel**

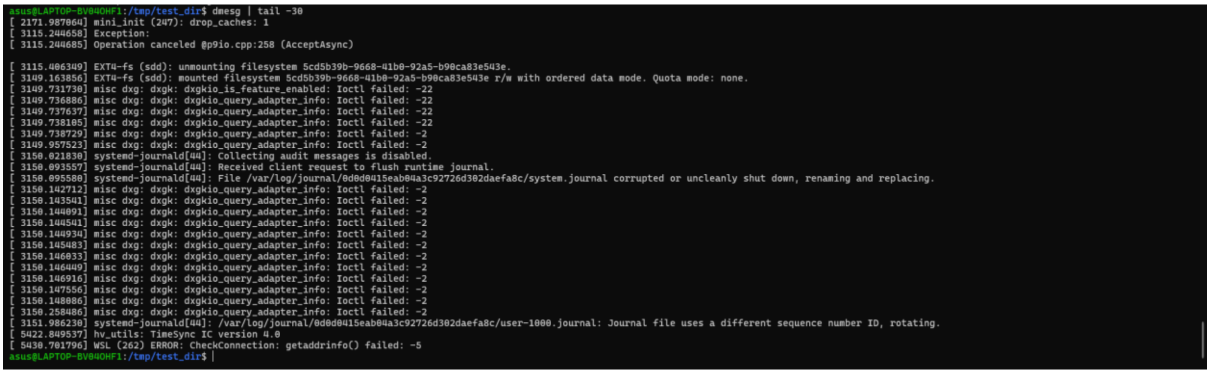


```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ uname -r
6.6.87.2-microsoft-standard-WSL2
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat parameter kernel**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
```

- **Lihat kernel ring buffer**



```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ dmesg | tail -30
[ 2171.987064] mini_init (247): drop_caches: 1
[ 3115.244658] Exception:
[ 3115.244685] Operation canceled @p9io.cpp:258 (AcceptAsync)

[ 3115.406349] EXT4-fs (sdd): unmounting filesystem 5cd5b39b-9668-41b0-92a5-b90ca83e543e.
[ 3149.163856] EXT4-fs (sdd): mounted filesystem 5cd5b39b-9668-41b0-92a5-b90ca83e543e r/w with ordered data mode. Quota mode: none.
[ 3149.731730] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 3149.736886] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.737637] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.738105] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.738729] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3149.957523] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.021830] systemd-journald[44]: Collecting audit messages is disabled.
[ 3150.093557] systemd-journald[44]: Received client request to flush runtime journal.
[ 3150.095580] systemd-journald[44]: File /var/log/journal/0d0d0415eab04a3c92726d302daefa8c/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 3150.142712] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.143541] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144091] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144541] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144934] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.145483] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146033] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146449] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.147556] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.148086] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.258486] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3151.986230] systemd-journald[44]: /var/log/journal/0d0d0415eab04a3c92726d302daefa8c/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 5422.849537] hv_utils: TimeSync IC version 4.0
[ 5430.701796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -5
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat kernel modules**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ lsmod
Module                  Size  Used by
tls                    98304  0
intel_rapl_msr         16384  0
intel_rapl_common      32768  1 intel_rapl_msr
kvm_amd               126976  0
kvm                   970752  1 kvm_amd
irqbypass              12288  1 kvm
crc32c_intel           16384  2
battery                20480  0
ac                     16384  0
sch_fq_codel           16384  1
configfs               53248  1
autofs4                45056  0
br_netfilter           28672  0
bridge                282624  1 br_netfilter
stp                    12288  1 bridge
llc                    12288  2 bridge,stp
ip_tables              28672  0
tun                    53248  0
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Informasi modul tertentu**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ modinfo ext4
name:           ext4
filename:       (builtin)
softdep:        pre: crc32c
license:        GPL
file:           fs/ext4/ext4
description:    Fourth Extended Filesystem
author:         Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias:          fs-ext4
alias:          ext3
alias:          fs-ext3
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

2.8 Operating-System Structure

Struktur sistem operasi (**Operating-System Structure**) terdiri dari beberapa lapisan yang saling berhubungan, mulai dari **hardware**, **kernel**, **system calls**, **system programs**, hingga **user applications**. Lapisan hardware berisi perangkat fisik seperti CPU dan USB, yang bisa dieksplor dengan perintah lspci, lsusb, dan cat /proc/cpuinfo. Kernel berperan sebagai penghubung utama antara perangkat keras dan perangkat lunak, dapat dilihat dengan uname -a. Lapisan **system calls** seperti strace ls memungkinkan aplikasi berkomunikasi dengan kernel. Lapisan **system programs** menyediakan alat bantu seperti ps aux dan top, sedangkan **user applications** seperti bash dan gcc berada di lapisan tertinggi.

Selain itu, praktik membandingkan **monolithic kernel** yang memuat semua layanan inti dalam satu sistem besar dengan **microkernel** yang lebih modular dan efisien. Program procfs_read.c digunakan untuk membaca informasi CPU, memori, dan proses melalui direktori virtual /proc, menunjukkan bagaimana Linux mengorganisasi dan menyediakan akses terstruktur terhadap komponen sistem operasi.

2.8.1 Eksplorasi Struktur Layering

- **Layer 1: Hardware**

```
asus@LAPTOP-BV04OHF1: /tn
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ lspci
1702:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
5582:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
db6e:00:00.0 System peripheral: Red Hat, Inc. Virtio file system (rev 01)
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ lsusb
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat /proc/cpuinfo
processor       : 0
vendor_id       : AuthenticAMD
cpu family      : 25
model           : 68
model name      : AMD Ryzen 7 7435HS
stepping        : 1
microcode       : 0xffffffff
cpu MHz         : 3094.099
cache size      : 512 KB
physical id     : 0
siblings        : 16
core id         : 0
cpu cores       : 8
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 13
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nons
top_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw
 topoext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrip_save tsc_scal
e vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
bugs            : sysret_ss_attrs null_seg spectre_v1 spectre_v2 spec_store_bypass srso ibpb_no_ret
bogomips        : 6188.19
TLB size        : 2560 4K pages
clflush size    : 64
cache_alignment : 64
address sizes   : 48 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id       : AuthenticAMD
cpu family      : 25
model           : 68
model name      : AMD Ryzen 7 7435HS
stepping        : 1
microcode       : 0xffffffff
cpu MHz         : 3094.099
cache size      : 512 KB
physical id     : 0
siblings        : 16
core id         : 0
cpu cores       : 8
apicid          : 1
initial apicid  : 1
fpu             : yes
fpu_exception   : yes
cpuid level     : 13
wp              : yes
```

- **Layer 2: Kernel**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ uname -a
Linux LAPTOP-BV04OHF1 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat /proc/sys/kernel/ostype
Linux
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Layer 3: System Calls**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffd64f01ae0 /* 26 vars */) = 0
brk(NULL)                               = 0x58f6eb1ba000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x70b50d0ef000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19935, ...}) = 0
mmap(NULL, 19935, PROT_READ, MAP_PRIVATE, 3, 0) = 0x70b50d0ea000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70b50d0bd000
mmap(0x70b50d0c3000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x70b50d0c3000
mmap(0x70b50d0e0000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x70b50d0e0000
mmap(0x70b50d0e6000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x70b50d0e6000
mmap(0x70b50d0e8000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x70b50d0e8000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70b50ce00000
mmap(0x70b50ce28000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x70b50ce28000
mmap(0x70b50cfb0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x70b50cfb0000
mmap(0x70b50cfff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x70b50cfff000
mmap(0x70b50d005000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x70b50d005000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70b50d023000
mmap(0x70b50d025000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x70b50d025000
mmap(0x70b50d093000, 163840, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x70b50d093000
mmap(0x70b50d0bb000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x97000) = 0x70b50d0bb000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x70b50d020000
arch_prctl(ARCH_SET_FS, 0x70b50d020800) = 0
set_tid_address(0x70b50d020ad0)         = 1418
set_robust_list(0x70b50d020ae0, 24)     = 0
rseq(0x70b50d021120, 0x20, 0, 0x53053053) = 0
mprotect(0x70b50cfff000, 16384, PROT_READ) = 0
mprotect(0x70b50d0bb000, 4096, PROT_READ) = 0
mprotect(0x70b50d0e6000, 4096, PROT_READ) = 0
mprotect(0x58f6b2347000, 8192, PROT_READ) = 0
mprotect(0x70b50d127000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x70b50d0ea000, 19935)           = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x4cf272b8, 0xfb167d3e]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_N
OSUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffdcd325390)      = -1 ENOENT (No such file or directory)
getrandom("\xe6\xaf\xca\x4b\x72\x81\x11\x21", 8, GRND_NONBLOCK) = 8
brk(NULL)                               = 0x58f6eb1ba000
brk(0x58f6eb1db000)                     = 0x58f6eb1db000
```

- **Layer 4: System Programs**

```
asus@LAPTOP-BV04OHF1: /tn
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  22008 12236 ?        Ss   Oct13   0:02 /sbin/init
root           2  0.0  0.0   3072  1664 ?        Sl   Oct13   0:00 /init
root           7  0.0  0.0   3072  1792 ?        Sl   Oct13   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          44  0.0  0.2  66816 17140 ?        S<s  Oct13   0:00 /usr/lib/systemd/systemd-journald
root          91  0.0  0.0  24876  6016 ?        Ss   Oct13   0:01 /usr/lib/systemd/systemd-udevd
systemd+     111  0.0  0.1  21456 12672 ?        Ss   Oct13   0:00 /usr/lib/systemd/systemd-resolved
systemd+     134  0.0  0.0  91024  7552 ?        Ssl  Oct13   0:00 /usr/lib/systemd/systemd-timesyncd
root         160  0.0  0.0   4236  2560 ?        Ss   Oct13   0:00 /usr/sbin/cron -f -P
message+     161  0.0  0.0   9596  4864 ?        Ss   Oct13   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root         168  0.0  0.1  17964  8320 ?        Ss   Oct13   0:00 /usr/lib/systemd/systemd-logind
root         170  0.0  0.1 1756096 12800 ?       Ssl  Oct13   0:00 /usr/libexec/wsl-pro-service -vv
root         173  0.0  0.0   3160  1920 hvc0     Ss+  Oct13   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
syslog       178  0.0  0.0 222508  5120 ?        Ssl  Oct13   0:00 /usr/sbin/rsyslogd -n -iNONE
root         192  0.0  0.0   3116  1792 tty1     Ss+  Oct13   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root         193  0.0  0.2 107032 22272 ?        Ssl  Oct13   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         282  0.0  0.0   3076   896 ?        Ss   Oct13   0:00 /init
root         283  0.0  0.0   3092  1152 ?        S    Oct13   0:00 /init
asus         284  0.0  0.0   6072  5120 pts/0    Ss   Oct13   0:00 -bash
root         285  0.0  0.0   6816  4352 pts/1    Ss   Oct13   0:00 /bin/login -f
asus         328  0.0  0.1  20312 11264 ?        Ss   Oct13   0:00 /usr/lib/systemd/systemd --user
asus         329  0.0  0.0  21152  3520 ?        S    Oct13   0:00 (sd-pam)
asus         346  0.0  0.0   6072  5120 pts/1    S+   Oct13   0:00 -bash
asus         408  0.0  0.0   2680  1408 pts/0    T    Oct13   0:00 ./simple_shell
root         665  0.0  0.0  14148  6784 pts/0    T    Oct13   0:00 sudo journalctl -u my_daemon
root         666  0.0  0.0  14148  2604 pts/2    Ss   Oct13   0:00 sudo journalctl -u my_daemon
root         667  0.0  0.0  17876  7704 pts/2    T+   Oct13   0:00 journalctl -u my_daemon
root         668  0.0  0.0   3512  2304 pts/2    S+   Oct13   0:00 less
root         909  0.0  0.0   3076  1028 ?        Ss   Oct13   0:00 /init
root         911  0.0  0.0   3092  1160 ?        S    Oct13   0:00 /init
asus         916  0.0  0.0   6072  5248 pts/3    Ss   Oct13   0:00 -bash
root         950  0.0  0.0  14152  6784 pts/3    S+   00:00   0:00 sudo nano Makefile
root         951  0.0  0.0  14152  2476 pts/4    Ss   00:00   0:00 sudo nano Makefile
root         952  0.0  0.0   4608  3712 pts/4    S+   00:00   0:00 nano Makefile
root        1280  0.0  0.2 370084 20352 ?        Ssl  00:15   0:00 /usr/libexec/packagekitd
polkitd     1285  0.0  0.0 308164  7808 ?        Ssl  00:15   0:00 /usr/lib/polkit-1/polkitd --no-debug
asus        1427  0.0  0.0   8280  4096 pts/0    R+   00:18   0:00 ps aux
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ top
top - 00:18:39 up  3:20,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  36 total,   1 running,  32 sleeping,   3 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7848.2 total,   7224.6 free,    561.5 used,    217.7 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7286.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   22008  12236   9292 S   0.0   0.2   0:02.47 systemd
      2 root      20   0    3072   1664   1664 S   0.0   0.0   0:00.01 init-systemd(Ub
      7 root      20   0    3072   1792   1792 S   0.0   0.0   0:00.00 init
     44 root      19  -1   66816  17140  16244 S   0.0   0.2   0:00.84 systemd-journal
     91 root      20   0   24876   6016   4864 S   0.0   0.1   0:01.25 systemd-udevd
    111 systemd+  20   0   21456  12672  10496 S   0.0   0.2   0:00.15 systemd-resolve
    134 systemd+  20   0   91024   7552   6784 S   0.0   0.1   0:00.37 systemd-timesyn
    160 root      20   0    4236   2560   2432 S   0.0   0.0   0:00.03 cron
    161 message+  20   0    9596   4864   4352 S   0.0   0.1   0:00.40 dbus-daemon
    168 root      20   0   17964   8320   7424 S   0.0   0.1   0:00.21 systemd-logind
    170 root      20   0 1756096  12800  10240 S   0.0   0.2   0:00.39 wsl-pro-service
    173 root      20   0    3160   1920   1792 S   0.0   0.0   0:00.01 agetty
    178 syslog    20   0  222508   5120   4480 S   0.0   0.1   0:00.20 rsyslogd
    192 root      20   0    3116   1792   1664 S   0.0   0.0   0:00.00 agetty
    193 root      20   0  107032  22272  13184 S   0.0   0.3   0:00.09 unattended-upgr
    282 root      20   0    3076    896    896 S   0.0   0.0   0:00.00 SessionLeader
    283 root      20   0    3092   1152   1024 S   0.0   0.0   0:00.63 Relay(284)
    284 asus      20   0    6072   5120   3456 S   0.0   0.1   0:00.23 bash
```

- **Layer 5: User Applications**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ which bash
/usr/bin/bash
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ which gcc
/usr/bin/gcc
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

2.8.2 Microkernel vs Monolithic

- **Lihat ukuran kernel (Monolithic)**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
```

- **Lihat built-in vs module**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat loaded modules**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ lsmod | wc -l
19
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

2.8.3 Program - Virtual File System

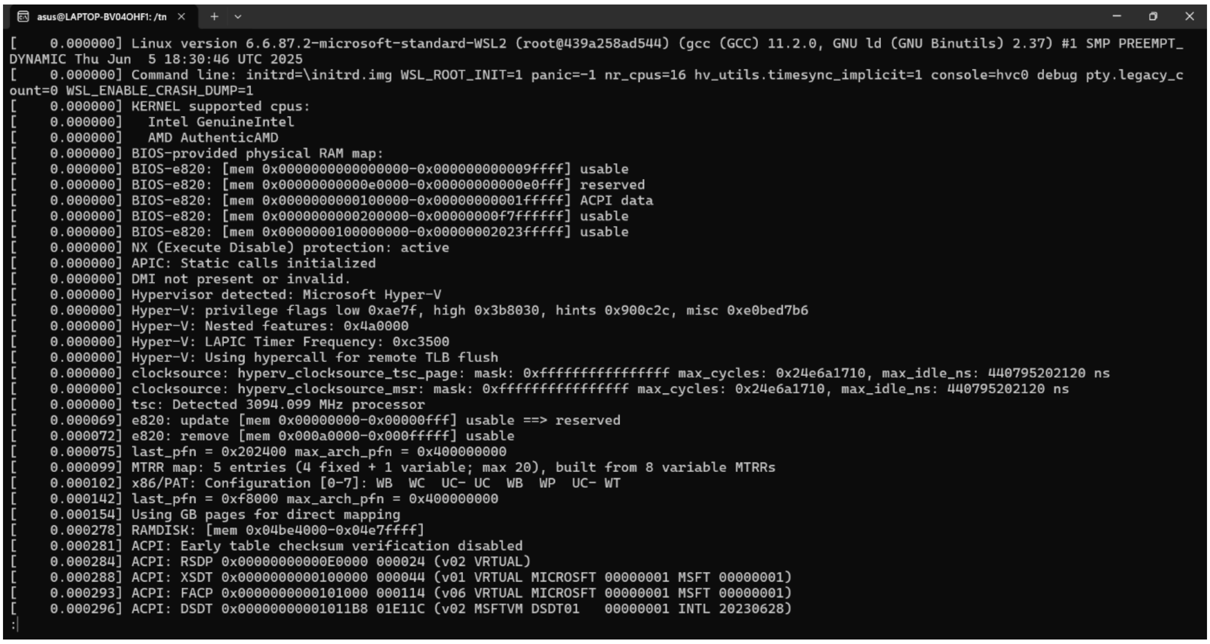
```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano procfs_read.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -o procfs_read procfs_read.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ./procfs_read
=== CPU Information ===
processor       : 0
vendor_id       : AuthenticAMD
cpu family      : 25
model           : 68
model name      : AMD Ryzen 7 7435HS
stepping        : 1
microcode       : 0xffffffff
cpu MHz         : 3094.099
cache size      : 512 KB
physical id     : 0

=== Memory Information ===
MemTotal:        8036548 kB
MemFree:         7365416 kB
MemAvailable:    7456148 kB
Buffers:            3924 kB
Cached:           215044 kB

=== Process Information ===
Name:   procfs_read
Umask:  0022
State:  R (running)
Tgid:   1503
Ngid:   0
Pid:    1503
PPid:   284
TracerPid:      0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

2.9 Building and Booting an Operating System

Pada praktik Building and Booting an Operating System, peserta mempelajari proses penting dalam menjalankan sistem Linux sejak komputer dinyalakan hingga sistem siap digunakan. Tahapan dimulai dari BIOS/UEFI yang mengeksekusi bootloader GRUB, yang bertugas memuat kernel Linux dan initramfs ke memori. Kernel kemudian menginisialisasi perangkat keras dan memulai proses utama sistem seperti systemd. Perintah dmesg digunakan untuk melihat pesan kernel selama booting, sementara systemd-analyze membantu menganalisis waktu dan urutan layanan saat startup. File /boot/grub/grub.cfg berisi konfigurasi GRUB yang menentukan opsi boot. Selain itu, parameter kernel yang aktif dapat dilihat dengan cat /proc/cmdline, dan dapat diubah sementara melalui menu GRUB atau secara permanen dengan mengedit file /etc/default/grub lalu menjalankan sudo update-grub. Praktik ini membantu memahami struktur dasar dan mekanisme kerja sistem operasi dari awal proses boot hingga sistem siap digunakan.

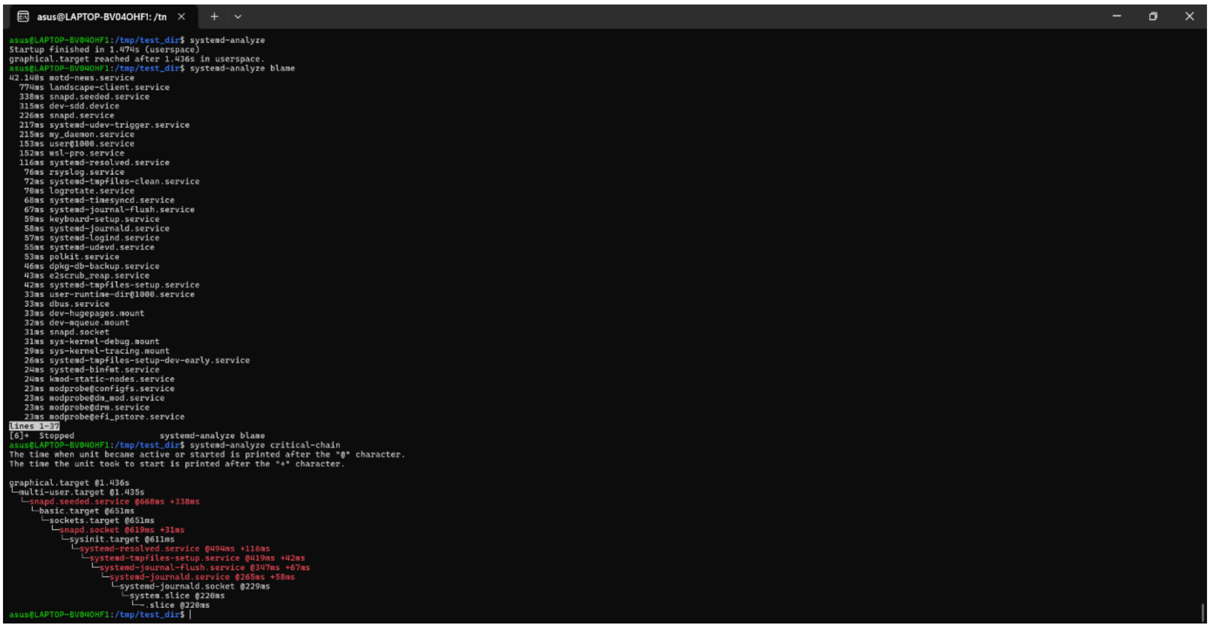
2.9.1 Analisis Boot Process

- **Lihat boot messages**



```
asus@LAPTOP-BV04OHF1: /tn
[    0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[    0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[    0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[    0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[    0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000f7ffffff] usable
[    0.000000] BIOS-e820: [mem 0x0000000100000000-0x00000002023fffff] usable
[    0.000000] NX (Execute Disable) protection: active
[    0.000000] APIC: Static calls initialized
[    0.000000] DMI not present or invalid.
[    0.000000] Hypervisor detected: Microsoft Hyper-V
[    0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
[    0.000000] Hyper-V: Nested features: 0x4a0000
[    0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[    0.000000] Hyper-V: Using hypercall for remote TLB flush
[    0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] tsc: Detected 3094.099 MHz processor
[    0.000069] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[    0.000072] e820: remove [mem 0x000a0000-0x000fffff] usable
[    0.000075] last_pfn = 0x202400 max_arch_pfn = 0x400000000
[    0.000099] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[    0.000102] x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
[    0.000142] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[    0.000154] Using GB pages for direct mapping
[    0.000278] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[    0.000281] ACPI: Early table checksum verification disabled
[    0.000284] ACPI: RSDP 0x00000000000E0000 000024 (v02 VRTUAL)
[    0.000288] ACPI: XSDT 0x0000000000100000 000044 (v01 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000293] ACPI: FACP 0x0000000000101000 000114 (v06 VRTUAL MICROSFT 00000001 MSFT 00000001)
[    0.000296] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01   00000001 INTL 20230628)
:
```

- **Lihat systemd boot analysis**



```
asus@LAPTOP-BV04OHF1: /tn
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ systemd-analyze
Startup finished in 1.474s (userspace)
graphical.target reached after 1.436s in userspace.
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ systemd-analyze blame
42.140s motd-news.service
  774ms landscape-client.service
  338ms snapd.seeded.service
  315ms dev-sdd.device
  226ms snapd.service
  217ms systemd-udev-trigger.service
  215ms my_daemon.service
  153ms user@1000.service
  152ms wsl-pro.service
  116ms systemd-resolved.service
   76ms rsyslog.service
   72ms systemd-tmpfiles-clean.service
   70ms logrotate.service
   68ms systemd-timesyncd.service
   67ms systemd-journal-flush.service
   59ms keyboard-setup.service
   58ms systemd-journald.service
   57ms systemd-logind.service
   55ms systemd-udevd.service
   53ms polkit.service
   46ms dpkg-db-backup.service
   43ms e2scrub_reap.service
   42ms systemd-tmpfiles-setup.service
   33ms user-runtime-dir@1000.service
   33ms dbus.service
   33ms dev-hugepages.mount
   32ms dev-mqueue.mount
   31ms snapd.socket
   31ms sys-kernel-debug.mount
   29ms sys-kernel-tracing.mount
   26ms systemd-tmpfiles-setup-dev-early.service
   24ms systemd-binfmt.service
   24ms kmod-static-nodes.service
   23ms modprobe@configfs.service
   23ms modprobe@dm_mod.service
   23ms modprobe@drm.service
   23ms modprobe@efi_pstore.service
lines 1-37
[6]+  Stopped                 systemd-analyze blame
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @1.436s
└─multi-user.target @1.435s
  └─snapd.seeded.service @668ms +338ms
    └─basic.target @651ms
      └─sockets.target @651ms
        └─snapd.socket @619ms +31ms
          └─sysinit.target @611ms
            └─systemd-resolved.service @494ms +116ms
              └─systemd-tmpfiles-setup.service @419ms +42ms
                └─systemd-journal-flush.service @347ms +67ms
                  └─systemd-journald.service @265ms +58ms
                    └─systemd-journald.socket @229ms
                      └─system.slice @220ms
                        └─.slice @220ms
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat GRUB configuration**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ |
```

- **Lihat initramfs**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
/usr/bin/unmkinitramfs: 64: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 57: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 38: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
cpio: premature end of archive
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

2.9.2 Using strace

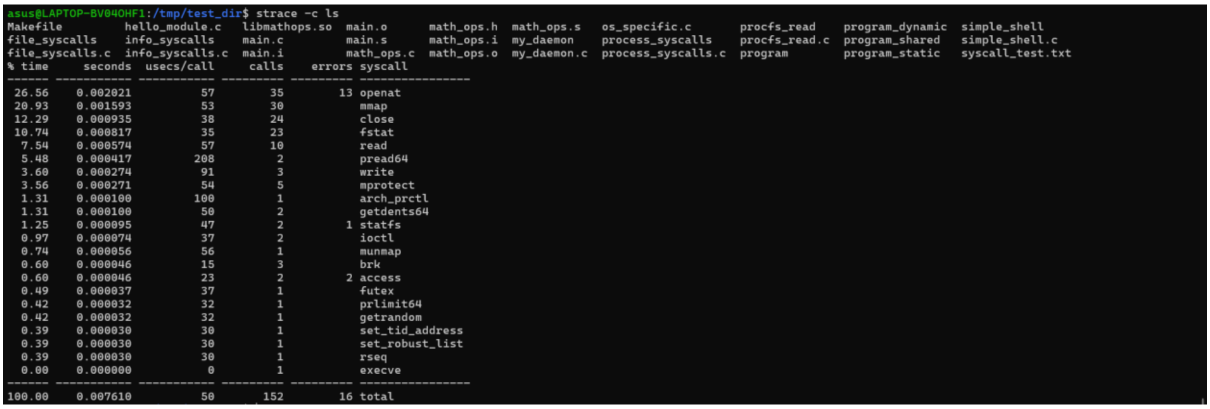
- **Trace system calls**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffc8acef790 /* 26 vars */) = 0
brk(NULL)                               = 0x60d86d44c000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x741d8c7f2000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19935, ...}) = 0
mmap(NULL, 19935, PROT_READ, MAP_PRIVATE, 3, 0) = 0x741d8c7ed000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x741d8c7c0000
mmap(0x741d8c7c6000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x741d8c7c6000
mmap(0x741d8c7e3000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x741d8c7e3000
mmap(0x741d8c7e9000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x741d8c7e9000
mmap(0x741d8c7eb000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x741d8c7eb000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x741d8c400000
mmap(0x741d8c428000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x741d8c428000
mmap(0x741d8c5b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x741d8c5b0000
mmap(0x741d8c5ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x741d8c5ff000
mmap(0x741d8c605000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x741d8c605000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x741d8c726000
mmap(0x741d8c728000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x741d8c728000
mmap(0x741d8c796000, 163840, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x741d8c796000
mmap(0x741d8c7be000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x97000) = 0x741d8c7be000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x741d8c723000
arch_prctl(ARCH_SET_FS, 0x741d8c723800) = 0
set_tid_address(0x741d8c723ad0)         = 1578
set_robust_list(0x741d8c723ae0, 24)     = 0
rseq(0x741d8c724120, 0x20, 0, 0x53053053) = 0
mprotect(0x741d8c5ff000, 16384, PROT_READ) = 0
mprotect(0x741d8c7be000, 4096, PROT_READ) = 0
```

- **Trace dengan detail**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ strace -v ls
execve("/usr/bin/ls", ["ls"], ["SHELL=/bin/bash", "WSL2_GUI_APPS_ENABLED=1", "WSL_DISTRO_NAME=Ubuntu", "NAME=LAPTOP-BV04OHF1", "PWD=/tmp/test_dir", "LOGNAME=asus", "HOME=
/home/asus", "LANG=C.UTF-8", "WSL_INTEROP=/run/WSL/283_interop", "LS_COLORS=rs=0:di=01;34:ln=01;36"..., "WAYLAND_DISPLAY=wayland-0", "LESSCLOSE=/usr/bin/lesspipe %s %"...
, "TERM=xterm-256color", "LESSOPEN=| /usr/bin/lesspipe %s", "USER=asus", "DISPLAY=:0", "SHLVL=1", "XDG_RUNTIME_DIR=/run/user/1000/", "WSLENV=", "XDG_DATA_DIRS=/usr/local/
share:/"..., "PATH=/usr/local/sbin:/usr/local/"..., "DBUS_SESSION_BUS_ADDRESS=unix:pa"..., "HOSTTYPE=x86_64", "PULSE_SERVER=unix:/mnt/wslg/Puls"..., "OLDPWD=/tmp", "_=/us
r/bin/strace"]) = 0
brk(NULL)                               = 0x5abbd28e2000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7a177f810000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=55554, st_mode=S_IFREG|0644, st_nlink=1, st_uid=0, st_gid=0, st_blksize=4096, st_blocks=40, st_size=19935, st_atime=1760366639
 /* 2025-10-13T21:43:59.991984907+0700 */, st_atime_nsec=991984907, st_mtime=1760366639 /* 2025-10-13T21:43:59.976159399+0700 */, st_mtime_nsec=976159399, st_ctime=176036
6639 /* 2025-10-13T21:43:59.984072152+0700 */, st_ctime_nsec=984072152}) = 0
mmap(NULL, 19935, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7a177f80b000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=13037, st_mode=S_IFREG|0644, st_nlink=1, st_uid=0, st_gid=0, st_blksize=4096, st_blocks=344, st_size=174472, st_atime=17603551
81 /* 2025-10-13T18:33:01.130906529+0700 */, st_atime_nsec=130906529, st_mtime=1727872852 /* 2024-10-02T19:40:52+0700 */, st_mtime_nsec=0, st_ctime=1759846701 /* 2025-10-
07T21:18:21.398866291+0700 */, st_ctime_nsec=398866291}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a177f7de000
mmap(0x7a177f7e4000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x7a177f7e4000
mmap(0x7a177f801000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x7a177f801000
mmap(0x7a177f807000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x7a177f807000
mmap(0x7a177f809000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7a177f809000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=39744, st_mode=S_IFREG|0755, st_nlink=1, st_uid=0, st_gid=0, st_blksize=4096, st_blocks=4152, st_size=2125328, st_atime=176035
5181 /* 2025-10-13T18:33:01.134906529+0700 */, st_atime_nsec=134906529, st_mtime=1758120942 /* 2025-09-17T21:55:42+0700 */, st_mtime_nsec=0, st_ctime=1759850153 /* 2025-1
0-07T22:15:53.856586392+0700 */, st_ctime_nsec=856586392}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7a177f400000
mmap(0x7a177f428000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7a177f428000
mmap(0x7a177f5b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x7a177f5b0000
mmap(0x7a177f5ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x7a177f5ff000
mmap(0x7a177f605000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7a177f605000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_dev=makedev(0x8, 0x30), st_ino=13002, st_mode=S_IFREG|0644, st_nlink=1, st_uid=0, st_gid=0, st_blksize=4096, st_blocks=1224, st_size=625344, st_atime=1760355
181 /* 2025-10-13T18:33:01.138906529+0700 */, st_atime_nsec=138906529, st_mtime=1727872851 /* 2024-10-02T19:40:51+0700 */, st_mtime_nsec=0, st_ctime=1759846701 /* 2025-10
```

- **Hitung waktu system calls**



```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ strace -c ls
Makefile          hello_module.c    libmathops.so  main.o       math_ops.h  math_ops.s    os_specific.c        procfs_read    program_dynamic  simple_shell
file_syscalls     info_syscalls     main.c         main.s       math_ops.i  my_daemon     process_syscalls     procfs_read.c  program_shared   simple_shell.c
file_syscalls.c   info_syscalls.c   main.i         math_ops.c   math_ops.o  my_daemon.c   process_syscalls.c   program        program_static   syscall_test.txt
% time     seconds  usecs/call     calls    errors syscall
------ ----------- ----------- --------- --------- ----------------
 26.56    0.002021          57        35        13 openat
 20.93    0.001593          53        30           mmap
 12.29    0.000935          38        24           close
 10.74    0.000817          35        23           fstat
  7.54    0.000574          57        10           read
  5.48    0.000417         208         2           pread64
  3.60    0.000274          91         3           write
  3.56    0.000271          54         5           mprotect
  1.31    0.000100         100         1           arch_prctl
  1.31    0.000100          50         2           getdents64
  1.25    0.000095          47         2         1 statfs
  0.97    0.000074          37         2           ioctl
  0.74    0.000056          56         1           munmap
  0.60    0.000046          15         3           brk
  0.60    0.000046          23         2         2 access
  0.49    0.000037          37         1           futex
  0.42    0.000032          32         1           prlimit64
  0.42    0.000032          32         1           getrandom
  0.39    0.000030          30         1           set_tid_address
  0.39    0.000030          30         1           set_robust_list
  0.39    0.000030          30         1           rseq
  0.00    0.000000           0         1           execve
------ ----------- ----------- --------- --------- ----------------
100.00    0.007610          50       152        16 total
```

- **Trace dan simpan ke file**

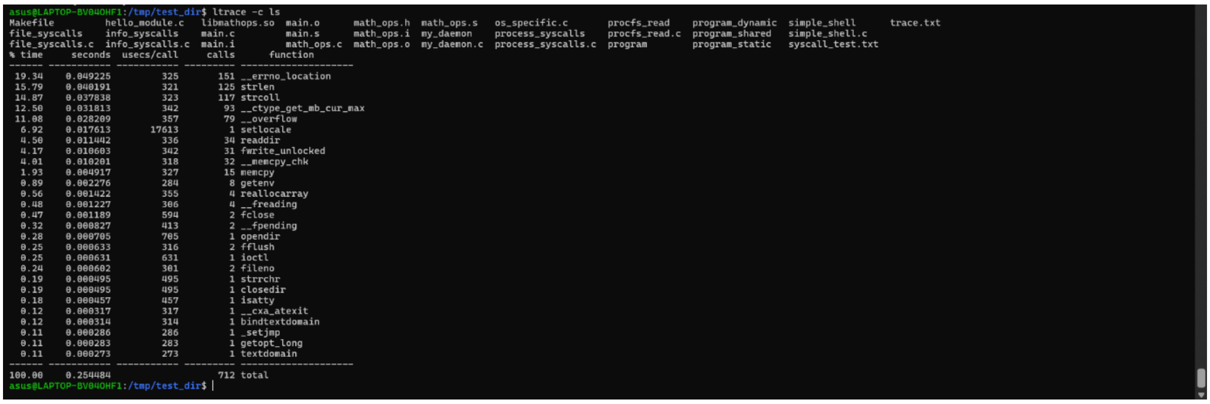
```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ strace -o trace.txt ls
Makefile         hello_module.c   libmathops.so  main.o      math_ops.h  math_ops.s    os_specific.c       procfs_read    program_dynamic  simple_shell     trace.txt
file_syscalls    info_syscalls    main.c         main.s      math_ops.i  my_daemon     process_syscalls    procfs_read.c  program_shared   simple_shell.c
file_syscalls.c  info_syscalls.c  main.i         math_ops.c  math_ops.o  my_daemon.c   process_syscalls.c  program        program_static   syscall_test.txt
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ cat trace.txt
execve("/usr/bin/ls", ["ls"], 0x7ffc0a5ef190 /* 26 vars */) = 0
brk(NULL)                               = 0x63fdd12cf000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x727e89821000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19935, ...}) = 0
mmap(NULL, 19935, PROT_READ, MAP_PRIVATE, 3, 0) = 0x727e8981c000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x727e897ef000
mmap(0x727e897f5000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x727e897f5000
mmap(0x727e89812000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x727e89812000
mmap(0x727e89818000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x727e89818000
mmap(0x727e8981a000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x727e8981a000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x727e89400000
mmap(0x727e89428000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x727e89428000
mmap(0x727e895b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x727e895b0000
mmap(0x727e895ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x727e895ff000
mmap(0x727e89605000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x727e89605000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x727e89755000
mmap(0x727e89757000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x727e89757000
mmap(0x727e897c5000, 163840, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x727e897c5000
mmap(0x727e897ed000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x97000) = 0x727e897ed000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x727e89752000
arch_prctl(ARCH_SET_FS, 0x727e89752800) = 0
set_tid_address(0x727e89752ad0)         = 1912
set_robust_list(0x727e89752ae0, 24)     = 0
```

2.9.3 Using ltrace

- **Trace library calls**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ltrace ls
strrchr("ls", '/')                                                              = nil
setlocale(LC_ALL, "")                                                           = "C.UTF-8"
bindtextdomain("coreutils", "/usr/share/locale")                                = "/usr/share/locale"
textdomain("coreutils")                                                         = "coreutils"
__cxa_atexit(0x55fce4908880, 0, 0x55fce4920008, 0)                              = 0
getopt_long(1, 0x7ffcdc479108, "abcdfghiklmnopqrstuvw:xABCDFGHI:"..., 0x55fce491f340, -1) = -1
getenv("LS_BLOCK_SIZE")                                                         = nil
getenv("BLOCK_SIZE")                                                            = nil
getenv("BLOCKSIZE")                                                             = nil
getenv("POSIXLY_CORRECT")                                                       = nil
getenv("BLOCK_SIZE")                                                            = nil
isatty(1)                                                                       = 1
ioctl(1, 21523, 0x7ffcdc478f20)                                                 = 0
getenv("TABSIZE")                                                               = nil
getenv("QUOTING_STYLE")                                                         = nil
__errno_location()                                                              = 0x7b59d64776a0
__errno_location()                                                              = 0x7b59d64776a0
getenv("TZ")                                                                    = nil
reallocarray(0, 100, 208, 0x55fd14c406c0)                                       = 0x55fd14c406d0
strlen(".")                                                                     = 1
__memcpy_chk(0x55fd14c45850, 0x55fce4917edc, 2, 2)                              = 0x55fd14c45850
__errno_location()                                                              = 0x7b59d64776a0
opendir(".")                                                                    = 0x55fd14c45870
readdir(0x55fd14c45870)                                                         = 0x55fd14c458a0
__errno_location()                                                              = 0x7b59d64776a0
__ctype_get_mb_cur_max()                                                        = 6
strlen("file_syscalls.c")                                                       = 15
strlen("file_syscalls.c")                                                       = 15
__memcpy_chk(0x55fd14c4d8b0, 0x55fd14c458b3, 16, 16)                            = 0x55fd14c4d8b0
readdir(0x55fd14c45870)                                                         = 0x55fd14c458c8
__errno_location()                                                              = 0x7b59d64776a0
__ctype_get_mb_cur_max()                                                        = 6
strlen("math_ops.i")                                                            = 10
strlen("math_ops.i")                                                            = 10
__memcpy_chk(0x55fd14c4d8d0, 0x55fd14c458db, 11, 11)                            = 0x55fd14c4d8d0
readdir(0x55fd14c45870)                                                         = 0x55fd14c458e8
__errno_location()                                                              = 0x7b59d64776a0
__ctype_get_mb_cur_max()                                                        = 6
strlen("process_syscalls.c")                                                    = 18
strlen("process_syscalls.c")                                                    = 18
__memcpy_chk(0x55fd14c4d8f0, 0x55fd14c458fb, 19, 19)                            = 0x55fd14c4d8f0
readdir(0x55fd14c45870)                                                         = 0x55fd14c45910
__errno_location()                                                              = 0x7b59d64776a0
__ctype_get_mb_cur_max()                                                        = 6
strlen("math_ops.o")                                                            = 10
strlen("math_ops.o")                                                            = 10
__memcpy_chk(0x55fd14c4d910, 0x55fd14c45923, 11, 11)                            = 0x55fd14c4d910
readdir(0x55fd14c45870)                                                         = 0x55fd14c45930
__errno_location()                                                              = 0x7b59d64776a0
__ctype_get_mb_cur_max()                                                        = 6
strlen("program_static")                                                        = 14
strlen("program_static")                                                        = 14
__memcpy_chk(0x55fd14c4d930, 0x55fd14c45943, 15, 15)                            = 0x55fd14c4d930
```

- **Dengan detail**



```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ ltrace -c ls
Makefile         hello_module.c   libmathops.so  main.o       math_ops.h  math_ops.s   os_specific.c        procfs_read    program_dynamic  simple_shell      trace.txt
file_syscalls    info_syscalls    main.c         main.s       math_ops.i  my_daemon    process_syscalls     procfs_read.c  program_shared   simple_shell.c
file_syscalls.c  info_syscalls.c  main.i         math_ops.c   math_ops.o  my_daemon.c  process_syscalls.c   program        program_static   syscall_test.txt
% time     seconds  usecs/call     calls      function
------ ----------- ----------- --------- --------------------
 19.34    0.049225         325       151 __errno_location
 15.79    0.040191         321       125 strlen
 14.87    0.037838         323       117 strcoll
 12.50    0.031813         342        93 __ctype_get_mb_cur_max
 11.08    0.028209         357        79 __overflow
  6.92    0.017613       17613         1 setlocale
  4.50    0.011442         336        34 readdir
  4.17    0.010603         342        31 fwrite_unlocked
  4.01    0.010201         318        32 __memcpy_chk
  1.93    0.004917         327        15 memcpy
  0.89    0.002276         284         8 getenv
  0.56    0.001422         355         4 reallocarray
  0.48    0.001227         306         4 __freading
  0.47    0.001189         594         2 fclose
  0.32    0.000827         413         2 __fpending
  0.28    0.000705         705         1 opendir
  0.25    0.000633         316         2 fflush
  0.25    0.000631         631         1 ioctl
  0.24    0.000602         301         2 fileno
  0.19    0.000495         495         1 strrchr
  0.19    0.000495         495         1 closedir
  0.18    0.000457         457         1 isatty
  0.12    0.000317         317         1 __cxa_atexit
  0.12    0.000314         314         1 bindtextdomain
  0.11    0.000286         286         1 _setjmp
  0.11    0.000283         283         1 getopt_long
  0.11    0.000273         273         1 textdomain
------ ----------- ----------- --------- --------------------
100.00    0.254484                   712 total
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

2.9.4 Using GDB

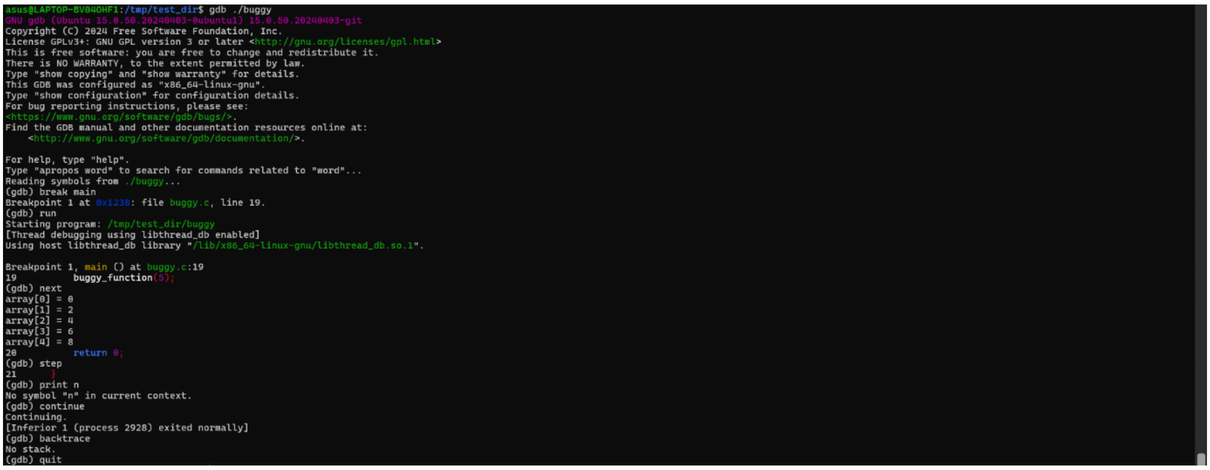
- **Debug dengan GDB**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ nano buggy.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

- **Compile dengan debug symbols**

```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gcc -g -o buggy buggy.c
asus@LAPTOP-BV04OHF1:/tmp/test_dir$
```

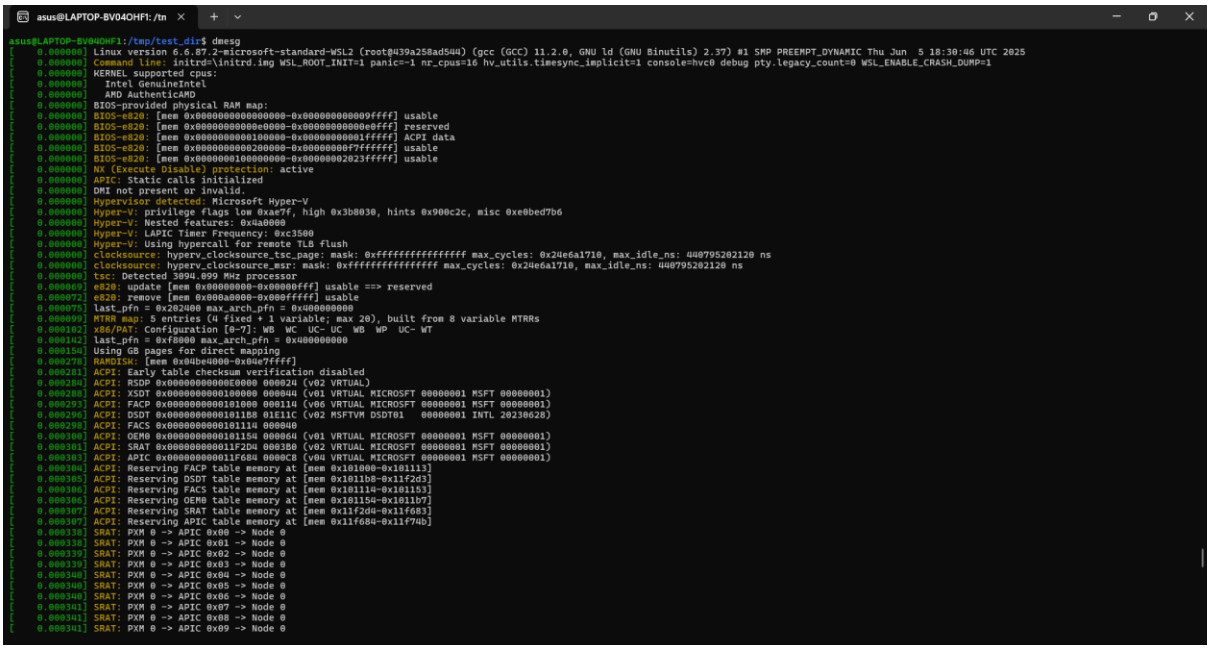
- **Di dalam GDB**



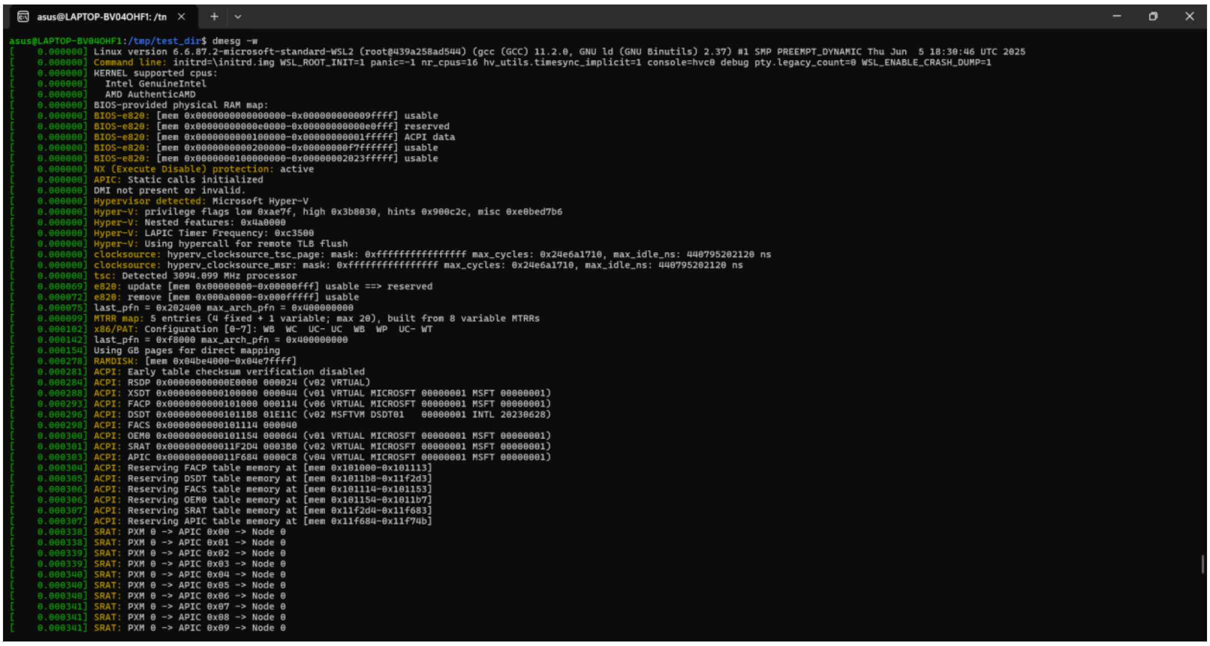
```
asus@LAPTOP-BV04OHF1:/tmp/test_dir$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 19.
(gdb) run
Starting program: /tmp/test_dir/buggy
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:19
19          buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
20          return 0;
(gdb) step
21      }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 2928) exited normally]
(gdb) backtrace
No stack.
(gdb) quit
```

2.9.5 Kernel Debugging dengan dmesg

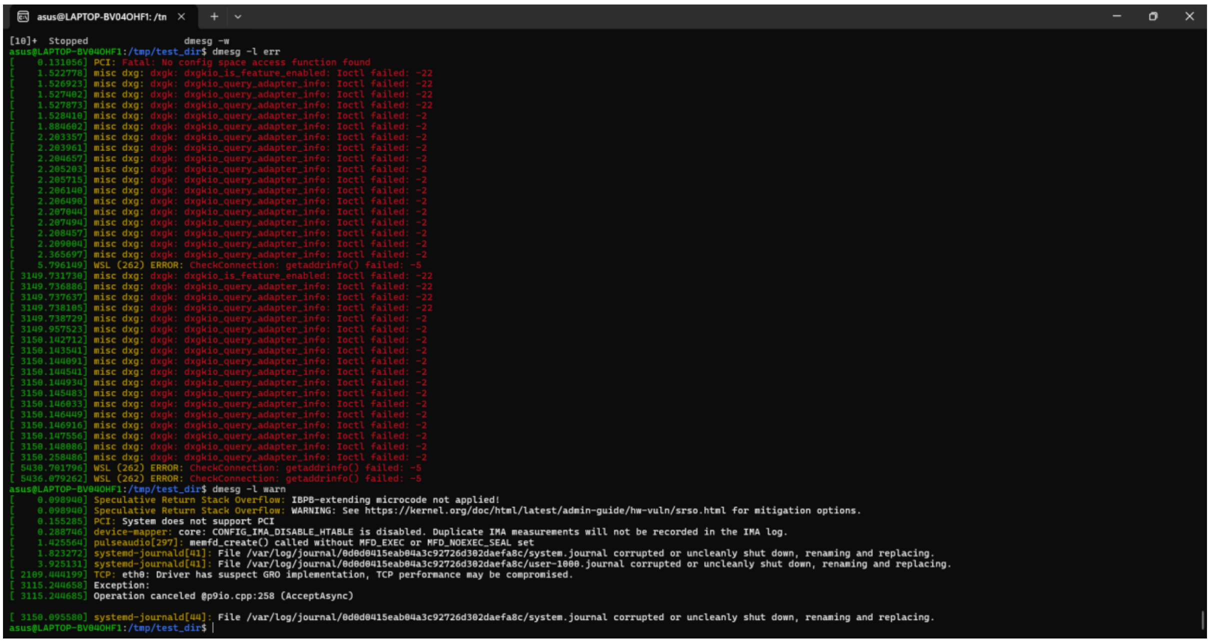
- **Lihat kernel log**



- **Follow kernel log**



- ○ **Filter by level**



2.9.6 Memory Leak Detection dengan Valgrind

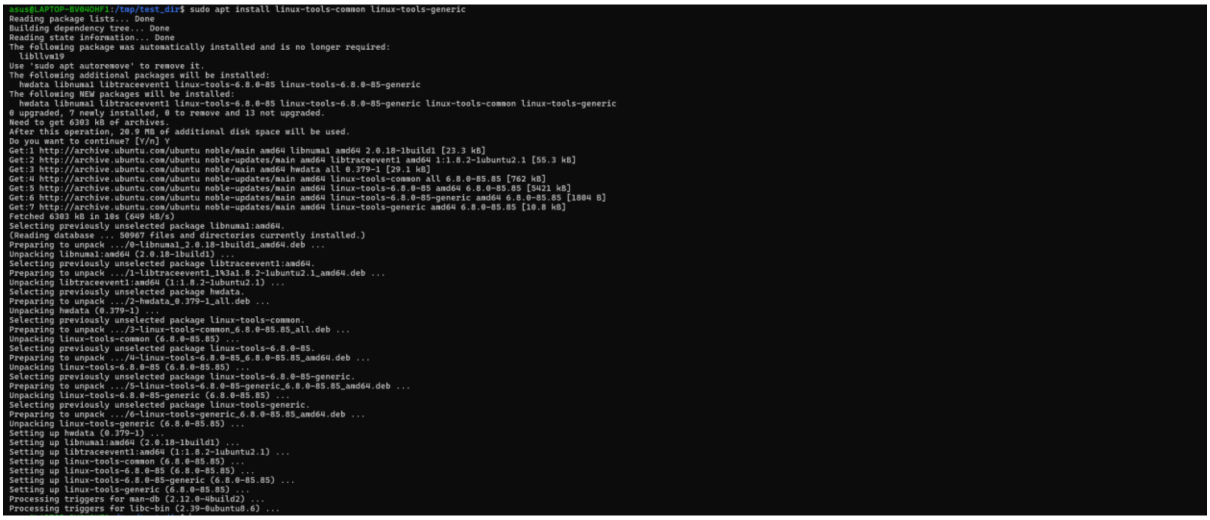
- **Install valgrind**

- **Buat program dengan memory leak**

- **Jalankan dengan valgrind**

2.9.7 Performance Analysis

- **CPU profiling dengan perf**



- **Monitor system calls**

- **Top-like interface**

BAB III

PENUTUP

KESIMPULAN

Dari hasil praktikum Sistem Operasi yang telah dilakukan, dapat disimpulkan bahwa sistem operasi memiliki peran utama sebagai pengelola seluruh sumber daya komputer dan penghubung antara perangkat keras serta perangkat lunak. Melalui berbagai percobaan pada lingkungan Linux, mahasiswa dapat memahami cara kerja sistem operasi dalam mengatur proses, memori, file, serta komunikasi antarproses melalui layanan seperti *system calls* dan *system services*.

Selain itu, mahasiswa juga mampu mengenal struktur sistem operasi, proses *linking* dan *loading*, serta interaksi antara pengguna, shell, dan kernel. Praktikum ini memberikan pengalaman langsung dalam mengimplementasikan teori melalui pemrograman C sederhana, sehingga memperkuat pemahaman terhadap konsep manajemen sistem operasi. Secara keseluruhan, kegiatan ini berhasil meningkatkan wawasan dan keterampilan mahasiswa dalam memahami mekanisme internal sistem operasi modern serta penerapannya dalam dunia komputasi.